Replication of the event segmentation findings by Panela et al. (2025, Communications Psychology)

Justin Bonagura Moran

2026-10-04

Table of contents

Introduction . . . 1
Methods . . . 3
Power Analysis . . . 3
Planned Sample . . . 3
Materials . . . 3
Procedure . . . 3
Analysis Plan . . . 3
Differences from Original Study . . . 4
Methods Addendum (Post Data Collection) . . . 4
Results . . . 4
Data preparation . . . 4
Confirmatory analysis . . . 4
Exploratory analyses . . . 4
Discussion . . . 5
Summary of Replication Attempt . . . 5
Commentary . . . 5

Introduction

Panela et al. (2025) asked whether large language models (LLMs) can sufficiently complete the task of event segmentation in narratives. Twenty participants read three written narratives from Trevor Noah's memoir Born a Crime, were asked to mark wherever they judged an event boundary to fall, and then recalled each narrative aloud. Then they had two LLMs (GPT-4 and LLaMA 3.0) perform the same event segmentation task, run twenty times per narrative

at three temperatures (0, 0.5, 1.0), so that the LLM sample size was the same as the human participant sample size. The target finding for this replication is the segmentation result. Researchers found that human event segmentation is more consistent with LLMs' segmentation than among humans themselves. The researchers split the participants into two groups of ten, and compared how many boundaries the two human groups shared against how many a human group shared with a group of LLM runs. The human-model overlap was larger, and it held at temperatures 0 and 0.5.

My current research asks how descriptions of people change as they pass between people, analyzing the compression and content convergence of social information across generations of transmission. My current version of this paradigm builds the seed stimuli for Generation 0 by sampling pieces of what real people said and creating an aggregated synthetic recall, so it needs a consistent and reliable way to slice a recall into chunks. Panela et al. do this by validating automated LLM event segmentation against humans performing the same task. However, Panela et al. run LLM event segmentation on a clean literary narrative, whereas my data is a set of messy spoken recalls of social information. Whether or not the method of event segmentation survives to character recall segmentation is not yet known, and I seek to explore this further in the future by building on this replication.

I plan to replicate the event segmentation portion of this study by running 42 online participants from UCSD's undergraduate participant pool (SONA). Participants will read a practice narrative and the three narratives presented in the original study, counterbalanced in order. The narratives will be presented via browser as one continuous block with no paragraph breaks and on a continuous scroll rather than fixed pages so that no page edge falls at the same word for every participant. Participants will digitally mark large event units for each of the three narratives. For LLMs, I will run Claude Opus 4.6 and gemma4 at temperatures 0, 0.5, and 1, with twenty runs per model per temperature per narrative to match the original study's sample size. Claude Opus 4.6 is chosen instead of a newer model because models newer than Opus 4.6 remove the temperature parameter, which is needed to run the three temperature conditions.

There will be four primary challenges. First, the practice narrative that calibrated participants on what a "large event" is, is not present in the public repository. I will request it from the authors. If I do not receive it in time, I will have to construct a calibration block myself. The original ran in person on printed paper, so moving the task to an online browser task slightly shifts reading modality and segmentation task modality. Excluding the recall task changes what participants are reading for, since the original's practice block told them to expect a recall task, and reading without that goal might produce different boundaries. Finally, the original models are not up to date, so this replication substitutes GPT-4 with Claude Opus 4.6 and LLaMA 3.0 with gemma4. Failure to reproduce results could be due to a change in models rather than a methodological failure.

Links

Repository: https://github.com/kyaneos/panela2025
Original paper: 2025_panela_llm_segmentation.pdf (Retrieved on Sep 30, 2026 from doi.org/10.1038/s44271-025-00359-7)

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.